



12-MAVZU · 42 TADAN · I BOB. HTML VA CSS

CSS animatsiyalar va transformatsiyalar

transform (translate, scale, rotate, skew), transition bilan silliq o'tishlar, @keyframes va animation xossalari; interaktiv tugmalar, yuklanish belgisi, kartalar effektlari.

2 soat · 1 nazariy + 1 amaliy

Bu darsda nimani o'rganasiz

- Elementlarni siljitish, kattalashtirish, burish (`transform`);
- Holat o'zgarganda silliq o'tish yaratish (`transition`);
- O'zi harakatlanadigan animatsiyalar (`@keyframes` + `animation`);
- Zamonaviy saytlardagi effektlar: "tirik" tugmalar, yuklanish belgisi, paydo bo'luvchi kartalar.

Transform — shaklni o'zgartirish

`transform` elementni **boshqa elementlarga ta'sir qilmasdan** siljitadi, o'lchamini o'zgartiradi yoki buradi.

HTML

```
<style>
  .qator { display: flex; gap: 30px; flex-wrap: wrap; padding: 30px; font-
family: sans-serif; }
  .q { width: 90px; height: 90px; background: #4f46e5; color: #fff; display:
grid; place-items: center;
    border-radius: 10px; font-size: 12px; text-align: center; }
  .t1 { transform: translateX(30px); } /* o'ngga 30px */
  .t2 { transform: translateY(-20px); } /* tepaga 20px */
  .t3 { transform: scale(1.3); } /* 1.3 barobar katta */
  .t4 { transform: scale(0.7); } /* kichraytirish */
  .t5 { transform: rotate(45deg); } /* 45 gradus burish */
  .t6 { transform: skew(15deg); } /* qiyshaytirish */
  .t7 { transform: rotate(-10deg) scale(1.1); } /* bir nechta birga */
</style>
<div class="qator">
  <div class="q">0ddiy</div>
  <div class="q t1">translateX</div>
  <div class="q t2">translateY</div>
  <div class="q t3">scale 1.3</div>
  <div class="q t4">scale 0.7</div>
  <div class="q t5">rotate 45°</div>
  <div class="q t6">skew</div>
  <div class="q t7">rotate + scale</div>
</div>
```

Funksiya	Vazifasi	Misol
<code>translate(x, y)</code>	Siljitish	<code>translate(10px, -5px)</code>
<code>scale(n)</code>	Kattalashtirish (1 = asl o'lcham)	<code>scale(1.1)</code> — 10% katta
<code>rotate(deg)</code>	Burish (gradus)	<code>rotate(180deg)</code> — teskari
<code>skew(deg)</code>	Qiyshaytirish	<code>skew(10deg)</code>



`transform-origin` — burish yoki kattalashtirish qaysi nuqtadan bo'lishi: `center` (standart), `top left`, `bottom`. Masalan, eshik ochilishi kabi effekt uchun `transform-origin: left`.

Tezlik egri chiziqlari

HTML

```
<style>
  .yolak { background: #f1f5f9; border-radius: 8px; margin: 8px 0; padding:
4px; font-family: sans-serif; font-size: 13px; }
  .shar { width: 120px; padding: 6px; background: #4f46e5; color: #fff; border-
radius: 20px; text-align: center;
    transition: transform 1.5s; }
  .sahna:hover .shar { transform: translateX(300px); }
  .linear .shar { transition-timing-function: linear; }
  .ease .shar { transition-timing-function: ease; }
  .ease-in .shar { transition-timing-function: ease-in; }
  .ease-out .shar { transition-timing-function: ease-out; }
  .ease-in-out .shar { transition-timing-function: ease-in-out; }
</style>
<div class="sahna">
  <p><b>Bu qutiga sichqonchani olib keling:</b></p>
  <div class="yolak linear"><div class="shar">linear</div></div>
  <div class="yolak ease"><div class="shar">ease</div></div>
  <div class="yolak ease-in"><div class="shar">ease-in</div></div>
  <div class="yolak ease-out"><div class="shar">ease-out</div></div>
  <div class="yolak ease-in-out"><div class="shar">ease-in-out</div></div>
</div>
```

- `linear` — bir tekis tezlikda (robotdek).
- `ease` — sekin boshlanib, tezlashib, sekin tugaydi (standart, tabiiy).
- `ease-in` — sekin boshlanadi; `ease-out` — sekin tugaydi.



`transition` ni **oddiy holatga** yozing (`.tugma`), `:hover` ga emas. Aks holda effekt faqat bir tomonga silliq bo'ladi — sichqoncha ketganda keskin qaytadi.

Animation — o'zi harakatlanuvchi effektlar

`transition` faqat holat o'zgarganda (hover, click) ishlaydi va faqat A dan B ga. **Animatsiya** esa o'zi boshlanadi, ko'p bosqichli bo'lishi va cheksiz takrorlanishi mumkin.

1-qadam: @keyframes — "ssenariy" yozish

CSS

```
@keyframes sakrash {  
  0%   { transform: translateY(0); }      /* boshlanish */  
  50%  { transform: translateY(-40px); }  /* o'rtasida — tepada */  
  100% { transform: translateY(0); }      /* oxiri — yana pastda */  
}  
/* Faqat ikki bosqich bo'lsa: from { ... } to { ... } */
```

2-qadam: elementga qo'llash

HTML

```
<style>
  @keyframes sakrash {
    0%, 100% { transform: translateY(0); }
    50%      { transform: translateY(-40px); }
  }
  @keyframes aylanish { to { transform: rotate(360deg); } }
  @keyframes yurak {
    0%, 100% { transform: scale(1); }
    15%      { transform: scale(1.25); }
    30%      { transform: scale(1); }
    45%      { transform: scale(1.15); }
  }
  @keyframes rang {
    0%   { background: #4f46e5; }
    33%  { background: #db2777; }
    66%  { background: #f59e0b; }
    100% { background: #4f46e5; }
  }

  .sahna { display: flex; gap: 40px; align-items: flex-end; padding: 60px 30px 20px; font-size: 48px; }
  .koptok { animation: sakrash 1s ease-in-out infinite; }
  .gildirak { animation: aylanish 2s linear infinite; }
  .yurak { animation: yurak 1.2s infinite; }
  .rangli { width: 80px; height: 80px; border-radius: 16px; animation: rang 4s infinite; }
</style>
<div class="sahna">
  <div class="koptok">⚽</div>
  <div class="gildirak">⚙️</div>
  <div class="yurak">❤️</div>
  <div class="rangli"></div>
</div>
```

CSS

```
animation: sakrash 1s ease-in-out 0s infinite alternate forwards;  
/* nomi | davomiyligi | tezlik | kechikish | takror soni | yo'nalish | oxirgi  
holat */
```

Xossa	Qiymatlar
animation-name	@keyframes nomi
animation-duration	2s , 500ms
animation-iteration-count	1 , 3 , infinite (cheksiz)
animation-direction	normal , reverse , alternate (borib-kelish)
animation-delay	0.5s — kechikib boshlash
animation-fill-mode	forwards — tugagach oxirgi holatda qolish

Amaliy misollar

1. Yuklanish belgisi (spinner)

HTML

```
<style>
.spinner {
  width: 50px; height: 50px;
  border: 5px solid #e0e7ff;
  border-top-color: #4f46e5;      /* faqat tepa qismi rangli */
  border-radius: 50%;
  animation: aylan 0.8s linear infinite;
  margin: 30px auto;
}
@keyframes aylan { to { transform: rotate(360deg); } }
</style>
<div class="spinner"></div>
<p style="text-align:center;font-family:sans-serif">Yuklanmoqda... </p>
```

2. Paydo bo'luvchi kartalar (ketma-ket)

HTML

```
<style>
  body { font-family: sans-serif; background: #f1f5f9; }
  @keyframes paydo {
    from { opacity: 0; transform: translateY(30px); }
    to   { opacity: 1; transform: translateY(0); }
  }
  .kartalar { display: grid; grid-template-columns: repeat(3, 1fr); gap: 16px;
padding: 20px; }
  .karta {
    background: white; padding: 24px; border-radius: 14px; box-shadow: 0 6px
18px rgba(0,0,0,.08);
    animation: paydo 0.6s ease both;
    transition: transform .25s, box-shadow .25s;
  }
  .karta:nth-child(2) { animation-delay: 0.15s; }
  .karta:nth-child(3) { animation-delay: 0.3s; }
  .karta:hover { transform: translateY(-8px); box-shadow: 0 16px 30px
rgba(79,70,229,.25); }
</style>
<div class="kartalar">
  <div class="karta">🍷 <b>HTML/CSS</b><br>Sahifa yaratish</div>
  <div class="karta">⚡ <b>JavaScript</b><br>Interaktivlik</div>
  <div class="karta">🐍 <b>Python</b><br>Backend</div>
</div>
<p style="text-align:center">"Ў Asl holiga" ni bosib, animatsiyani qayta
ko'ring</p>
```

3. Porlovchi tugma

HTML

```
<style>
  .porla {
    background: linear-gradient(90deg, #4f46e5, #db2777, #4f46e5);
    background-size: 200% 100%;
    color: white; border: none; padding: 16px 34px; font-size: 18px;
    border-radius: 999px; cursor: pointer; font-weight: bold;
    animation: oqim 3s linear infinite;
    transition: transform .2s;
  }
  .porla:hover { transform: scale(1.06); }
  .porla:active { transform: scale(.96); }
  @keyframes oqim { to { background-position: -200% 0; } }
</style>
<div style="padding:40px;text-align:center"><button class="porla">Kursga
yozilish 🚀</button></div>
```

Unumdorlik va foydalanuvchilar haqida qayg'urish

- Animatsiya uchun eng yaxshi xossalar — `transform` va `opacity`: ular tez ishlaydi. `width`, `top`, `margin` ni animatsiya qilish sahifani sekinlashtiradi.
- Animatsiyani **me'yorida** ishlatish: hamma narsa sakrab tursa, ko'z charchaydi.
- Ba'zi odamlarda harakatdan bosh aylanadi. Ular uchun: `@media (prefers-reduced-motion: reduce) { * { animation: none !important; transition: none !important; } }`

Amaliy mashg'ulot

- 1 Portfolio saytingizdagi barcha tugma va havolalarga `transition` bilan silliq `:hover` effektlari qo'shing.
- 2 Loyiha kartalari hover'da 8px tepaga ko'tarilsin va soyasi kuchaysin.
- 3 Hero bo'limidagi sarlavha va tugma sahifa ochilganda pastdan yuqoriga "paydo" bo'lsin (`@keyframes paydo` + kechikish).

- 4 Profil rasmingizga sekin "nafas olish" effekti: `scale(1)` → `scale(1.03)` → `scale(1)`, 4 soniya, cheksiz.
- 5 **Ijodiy topshiriq:** faqat CSS bilan "Quyosh tizimi" yarating: markazda quyosh 🌞, atrofida 2 ta sayyora turli tezlikda aylanadi. (Maslahat: sayyorani o'zidan katta shaffof doira ichiga qo'ying va doirani aylantiring.)

Ko'p uchraydigan xatolar

- ✗ Bir elementga ikkita `transform` yozish — ikkinchisi birinчисini o'chiradi. Bittada yozing:
`transform: rotate(10deg) scale(1.2);`
- ✗ `transition: all` ni hamma joyda ishlatish — kutilmagan xossalar ham animatsiya bo'ladi; aniq xossani ko'rsating.
- ✗ `@keyframes` nomi va `animation` dagi nom mos emas (masalan, `aylanish` va `aylanis`).
- ✗ `display: none` ni animatsiya qilishga urinish — u animatsiya bo'lmaydi; `opacity` va `visibility` ishlatish.

O'zingizni tekshiring

? Elementni 1.2 barobar kattalashtiruvchi kod?

- A. `transform: size(1.2)`
- B. `zoom: 120px`
- C. `transform: scale(1.2)`
- D. `transform: grow(20%)`

? **Hover effekti silliq bo'lishi uchun `transition` qayerga yoziladi?**

- A. Elementning oddiy holatiga
- B. Faqat `:hover` ga
- C. `@keyframes` ichiga
- D. `body` ga

? **Animatsiyani cheksiz takrorlash uchun qaysi qiymat?**

- A. `repeat`
- B. `loop`
- C. `forever`
- D. `infinite`

? **`transition` va `animation` ning asosiy farqi?**

- A. Farqi yo'q
- B. `transition` holat o'zgarganda ishlaydi; `animation` o'zi boshlanadi va ko'p bosqichli bo'lishi mumkin
- C. `animation` faqat ranglar uchun
- D. `transition` faqat telefonda ishlaydi

? Qaysi xossalarni animatsiya qilish eng tez (unumdor)?

- A. `width` va `height`
- B. `transform` va `opacity`
- C. `margin` va `padding`
- D. `top` va `left`



Uyga vazifa

1. "Quyosh tizimi" ijodiy topshirig'ini yakunlang.
2. 3 xil yuklanish belgisi (spinner) yarating: aylanuvchi doira, sakrovchi uchta nuqta, to'lib boruvchi chiziq.
3. Kartani 180° aylantirib, orqa tomonini ko'rsatuvchi "flip card" effektini yarating (`rotateY` , `backface-visibility` xossalari internetdan o'rganing).



Yodda tuting

- `transform`: `translate` , `scale` , `rotate` , `skew` — boshqa elementlarga ta'sir qilmaydi.
- `transition`: xossa vaqt tezlik; — oddiy holatga yoziladi, holat o'zgarishini silliq qiladi.
- `@keyframes nom { 0% {...} 100% {...} }` + `animation: nom 2s infinite;` — o'zi ishlaydigan animatsiya.
- Unumdorlik: `transform` va `opacity` . Me'yorni saqlang.

☐ Darsni o'qib bo'ldim

✓ Test javoblari

1. Elementni 1.2 barobar kattalashtiruvchi kod? — **C) transform: scale(1.2)**
2. Hover effekti silliq bo'lishi uchun transition qayerga yoziladi? — **A) Elementning oddiy holatiga**
3. Animatsiyani cheksiz takrorlash uchun qaysi qiymat? — **D) infinite**
4. transition va animation ning asosiy farqi? — **B) transition holat o'zgarganda ishlaydi; animation o'zi boshlanadi va ko'p bosqichli bo'lishi mumkin**
5. Qaysi xossalarni animatsiya qilish eng tez (unumdor)? — **B) transform va opacity**